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# SEXUAL DIFFERENCE IN THE MIGRATION PATTERN OF BLUE MARLIN, *MAKAIRA NIGRICANS*, RELATED TO SPAWNING AND FEEDING ACTIVITIES IN THE WESTERN AND CENTRAL NORTH PACIFIC OCEAN

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## ABSTRACT

The reproductive condition and stomach contents of blue marlin, *Makaira nigricans* Lacépède, 1802 (n = 645), were quantitatively investigated in three different regions of the North Pacific Ocean between 2003 and 2009. Males strongly dominated (females:males = 34:439) in Region III (4°N–21°N, 131°E–154°W) throughout the year, and eight females (28%) had ovaries in the maturing or spawning stage. Although the sampling months were limited to September–November in Region II (18°N–32°N, 171°W–140°W), the sex ratio was more similar (females:males = 28:26) and there was no evidence of spawning. Only females (n = 100) were observed in Region I (33°N–36°N, 135°E–140°E) from July to September, the main season when blue marlin occur off the coast of Japan, and no females had ovaries in the maturing or spawning stage. Stomach-content analysis revealed that the feeding intensity of females was higher in Region I than in the other two regions. These results suggest that blue marlin prey items may be more abundant at non-spawning areas in Region I, to which female blue marlin migrate for feeding. In contrast, lower feeding intensities and evidence of spawning in Region III suggest that blue marlin prey may be scarce in spawning areas. Male blue marlin tend to remain and wait for females in spawning areas, forgoing a feeding migration.

Some oceanic fish species do not strongly rely on coastal areas, enabling them to migrate extensively. Such species change their distribution range according to preferred water temperatures and/or to pursue their prey; e.g., Pacific bluefin tuna, *Thunnus orientalis* (Temminck and Schlegel, 1844), and swordfish, *Xiphias gladius* Linnaeus, 1758, migrate to higher latitude areas in the summer and lower latitude areas in the winter (Inagake et al. 2001, Takahashi et al. 2003). These migrations are sometimes strongly related to their spawning and feeding activities (e.g., Kuroda 1991, Ichii et al. 2009). Swordfish migrate north–south pursuing schools of neon flying squid, *Ommastrephes bartramii* (Lesueur, 1821), in the North Pacific (Watanabe et al. 2009). Blue shark, *Prionace glauca* Linnaeus, 1758, migrate southward for mating and northward for parturition in the North Pacific (Nakano 1994). Furthermore, southern bluefin tuna, *Thunnus maccoyii* (Castelnau, 1872), migrate to southern oceans for feeding and to specific northern areas for spawning (Farley et al. 2007).

The blue marlin, *Makaira nigricans* Lacépède, 1802, is a large migratory oceanic teleost with an extensive distribution range, spanning from tropical to temperate waters (45°N–45°S) of the Indo-Pacific and Atlantic Oceans (Nakamura 1985). Tagging data demonstrate that blue marlin can move from ocean to ocean (e.g., Atlantic-Indian or Indian-Pacific) throughout their life history (Ortiz et al. 2003), but tag-recapture data are quite limited to show their actual horizontal movement pattern.

There is only one known long term (up to ~1 yr) tracking study on blue marlin which logged their horizontal movement in the Gulf of Mexico by pop-up satellite archival tags (Kraus et al. 2011). Migration patterns of blue marlin are typically estimated using seasonal catch data of longline fisheries (Anraku and Yabuta 1959, Mather et al. 1972), and such patterns often relate to seasonal changes in sea-surface temperature (Su et al. 2008). Blue marlin migrate northward from April to August in the western North Pacific and southward after August in the central North Pacific (Anraku and Yabuta 1959). Differential distribution between the sexes has also been reported for blue marlin and may relate to reproductive behavior (Kume and Joseph 1969). The number of females is higher than that of males near the Galapagos Islands, but not in other areas of the eastern Pacific (Kume and Joseph 1969).

The spawning area of blue marlin in the Pacific Ocean has been previously estimated from larval-occurrence data (Nishikawa et al. 1985), from the gonad index of adult fish (Kume and Joseph 1969), and from combined analysis of these data (Nakamura 1983). Other studies have provided evidence of spawning around Hawaii through collections of fertilized eggs (Hyde et al. 2005) and in the western North Pacific through histological observations of ovaries (Shimose et al. 2009, Sun et al. 2009). According to the review by Nakamura (1983), the spawning area of blue marlin is restricted between 30°N and 25°S in the western and central Pacific, with extensive non-spawning regions (30°N–45°N, 25°S–45°S). Restricted spawning areas and seasonal migration patterns imply that blue marlin undergo a spawning migration. To document such spawning migrations, the reproductive condition of females and males must be examined over a wide geographic range.

Feeding activity is also thought to be related to the migration behavior of blue marlin. Although the diet of blue marlin has been well studied (Erdman 1962, Brock 1984, Abitia-Cardenas et al. 1999, Shimose et al. 2006), variation in feeding intensity across different areas has not yet been examined. The objectives of the present study were to compare the distribution of female and male blue marlin and to evaluate reproductive condition and feeding intensity across different latitudinal areas of the North Pacific Ocean. Our goal was to use these data to examine sexual differences in the migration pattern of blue marlin as related to spawning and feeding activities.

## Materials and Methods

Sample Collection and Measurements.—Blue marlin specimens were collected in the western and central North Pacific Ocean between 2003 and 2009 (Fig. 1). Sport fishing tournaments targeting marlin species are held only in the summer season (July–September) when blue marlin migrate to areas off the coast of Japan (Yatomi 1995). Blue marlin were caught by lure trolling during 1-d fishing trips and were also landed at four fishing ports (33°29´N–36°54´N, 135°46´E–140°47´E) in Japan. This fishing area is close to the northern limit of the blue marlin distribution in the western North Pacific Ocean (Nakamura 1985), and it is defined here as Region I.

Blue marlin caught by the longline training voyages of Japanese fisheries' high school vessels operated year round except for March and August. Their main target is bigeye tuna, *Thunnus obesus* (Lowe, 1839), but other tunas and billfishes are also landed and processed. The fishing area of these training vessels was broad and was divided into two major regions by geographic location and known spawning ground: Region II (18°42´N–32°42´N, 171°34´W–140°10´W) located north of the Hawaiian archipelago, and Region III (4°46´N–21°41´N, 131°29´E–154°41´W) located in a lower latitude area corresponding to the known spawning ground of blue marlin

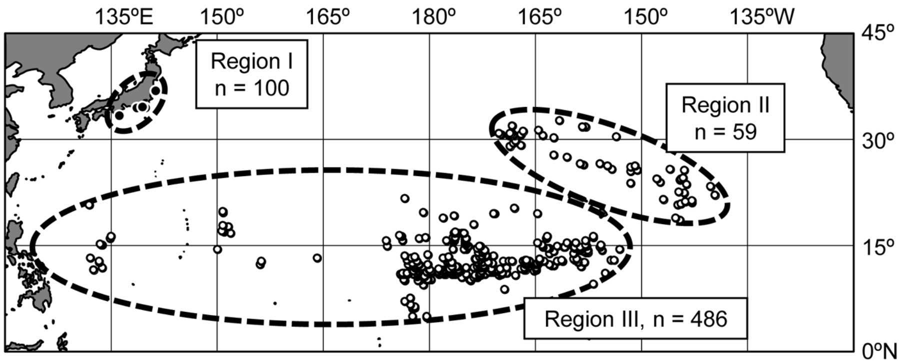


Figure 1. Sampling locations for blue marlin, *Makaira nigricans*, in the North Pacific Ocean. Open and closed circles indicate longline operation points and landing ports of sport fishing tournaments, respectively. Sampling locations were divided into three major regions.

(Nishikawa et al. 1985). Sea-surface temperatures were measured by the fishing vessels, and ranges were ~18–28, 23–27, and 25–30 °C in Regions I, II, and III, respectively. Annual data in both fishing tournaments (2003–2008) and longline training (2004–2009) were pooled for analyses.

Measurements of lower jaw–fork length (LJFL; to the nearest 1 cm) and processed weight (PW; body weight without bill, caudal fin, gills, and viscera; to the nearest 1 kg) were used in this study. When LJFL (n = 51) and PW (n = 421) were not recorded, the eye–fork length (EFL) or the whole body weight (BW) were later converted to LJFL and PW using LJFL–EFL (cm; $LJFL = 7.51 + 1.11 \times EFL$) and PW–BW (kg; $PW = -0.532 + 0.901 \times BW$) equations (Shimose et al. 2009). Sex was determined by visual inspection of gonad morphology. The smallest mature female is estimated to be 155 cm EFL (Kume and Joseph 1969) corresponding to 180 cm LJFL: and the smallest mature male is estimated to be 141 cm LJFL (see Results). Females smaller than 180 cm LJFL and males smaller than 140 cm were obviously juveniles (termed "juvenile" hereafter), and they were excluded for calculation of sex ratios.

Observations of the Gonads.—At the fishing tournaments (Region I), gonads were removed, and a portion was fixed in 10% buffered formalin for histological analyses. On the training vessels (Regions II and III), gonad samples were frozen. Frozen ovaries were thawed and fixed in 10% buffered formalin in the laboratory. Although frozen ovaries may not be appropriate for detailed histological observations, histological stages have been successfully identified for swordfish (Young et al. 2003). Formalin-fixed ovaries (n = 152) were dehydrated, embedded in paraffin wax, and then sectioned into 7–10-μm thick samples before staining with haematoxylin and eosin. Ovarian stages were identified based on the most advanced oocyte, with reference to previous studies (Yamamoto and Yamazaki 1961, Arocha 2002, Shimose et al. 2009). The stages were classified into four phases (Fig. 2): (1) immature, only chromatin nucleolus oocytes were observed; (2) inactive, only peri-nucleolus, yolk vesicle, or early yolked oocytes were observed; (3) maturing, fully yolked oocytes were observed; (4) spawning, migratory nucleus oocytes, hydrated oocytes, or postovulatory follicles were observed. Although atretic oocytes were observed in some ovaries, the ratios did not exceed 50%.

For males (collected in Regions II and III only), testes (n = 301) were defrosted and cut using a knife to check whether milt was exuded. Exudation of milt from the testes is strong evidence that the testes are in an active condition. In contrast, the absence of exuded milt indicates an inactive or post-spawning condition, with a smaller proportion of the current energy reserves dedicated to spawning.

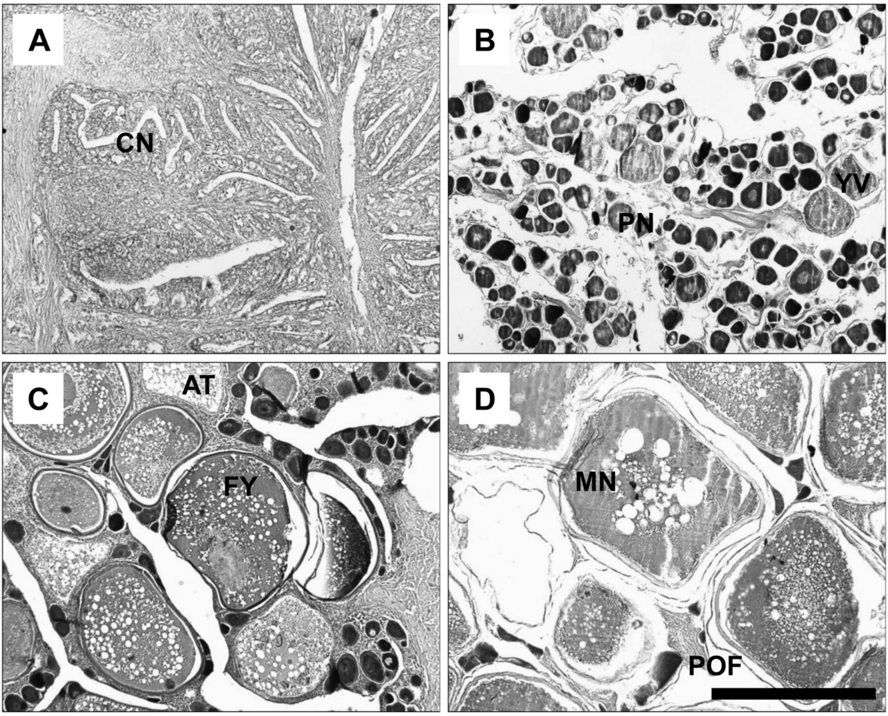


Figure 2. Photomicrographs of four ovarian stages of blue marlin, *Makaira nigricans:* (A) immature, 130 cm LJFL; (B) inactive, 257 cm; (C) maturing, 184 cm; (D) spawning, 223 cm. All ovaries were frozen before histological observation. CN: chromatin nucleolus oocyte; PN: peri-nucleolus oocyte; YV: yolk vesicle oocyte; FY: fully yolked oocyte; MN: migratory nucleus oocyte; AT: atretic oocyte; POF: postovulatory follicle. Scale bar = 500 μm.

Stomach Contents.—Stomach contents of blue marlin caught during the fishing tournaments were removed and returned to the laboratory on ice. Samples collected from training vessels were frozen at sea. Stomachs (n = 484) were initially inspected as to whether they were empty or contained prey items. Stomach contents were identified to the lowest possible taxon, counted, and weighed. The importance of each prey taxon was evaluated by the frequency of occurrence (%F, number of stomachs containing the prey taxon × 100 / number of stomachs containing any prey items), the number of prey (%N, number of the prey taxon × 100 / total number of all prey items), prey weight (%W, total weight of the prey taxon × 100 / total weight of all prey items), and the index of relative importance (%IRI, IRI of the prey taxon × 100 / total IRI of all prey items) using IRI = (%N + %W) × %F (Pinkas et al. 1971).

Each prey item was classified into one of five digestive classes: (1) undigested with skin color recognizable; (2) undigested with body surface damaged; (3) partially digested and most parts remaining; (4) nearly digested and only some parts remaining; (5) well digested and a few hard structures remaining. All items were weighed to the nearest 1 g. The body lengths of prey items in digestive classes 1 and 2 were measured to the nearest 1 mm, and those in class 3 were estimated to the nearest 1 cm. Mantle length, total length, and standard length were used for the body lengths of cephalopods, crustaceans, and teleosts, respectively.

The stomach content index (SCI) was calculated using stomach content weight (SCW) and processed weight (PW) using the following equation: SCI = SCW × 100 × $PW^{-1}$ (Shimose et al. 2006). This index is a simple criterion for evaluating fish repletion and can be used to assess

Table 1. Number of female and male blue marlin, *Makaira nigricans*, sampled by month (years, 2003–2009, combined). Only females were collected in Region I. Juveniles (females < 180 cm and males < 140 cm) were excluded. Sex ratios were compared to 1:1 using Chi-square test. –: not collected, **: P < 0.01, ns: not significant.

| | Region I | | Region II | | | Region III | | |
|---|---|---|---|---|---|---|---|---|
| Month | Female | Test | Female | Male | Test | Female | Male | Test |
| January | – | | – | – | | 2 | 55 | ** |
| February | – | | – | – | | 8 | 91 | ** |
| March | – | | – | – | | – | – | |
| April | – | | – | – | | 1 | 15 | ** |
| May | – | | – | – | | 14 | 199 | ** |
| June | – | | – | – | | 5 | 41 | ** |
| July | 50 | ** | – | – | | 1 | 3 | ns |
| August | 14 | ** | – | – | | – | – | |
| September | 36 | ** | 5 | 6 | ns | – | – | |
| October | – | | 15 | 14 | ns | 2 | 17 | ** |
| November | – | | 8 | 6 | ns | 1 | 17 | ** |
| December | – | | – | – | | 0 | 1 | ns |
| Total | 100 | ** | 28 | 26 | ns | 34 | 439 | ** |

prey abundance (Shimose et al. 2006, 2010, Watanabe et al. 2006). Excluding juvenile data, values of the mean SCI of female and male blue marlin were compared among regions.

Statistical Analyses.—Chi-square tests were applied to determine whether the sex ratio differed from 1:1. Mean values of LJFL were compared between sexes or between two different areas using Welch's t-tests, and pairwise t-tests adjusted by the Holm method were performed to compare LJFL among the three regions.

The digestive state of stomach contents can potentially differ among samples caught using different fishing gear. Thus, the composition of digestive classes 1–5 was compared between samples collected by the two types of fishing gear using Mann-Whitney U-tests. The lack of a significant difference ($P > 0.05$) indicated that two samples were in the same digestive condition and reflected similar feeding activities. This test was further applied to specific taxa (Teuthoidea and Thunnini) that were abundant in samples from both types of gear as well as to taxa that exhibited low morphological divergence.

Single prey weights or SCI were compared among areas using Mann-Whitney U-tests and pairwise U-tests adjusted by the Holm method. All analyses were performed using the program R (R Development Core Team 2008). Values are reported as means ± standard deviation (SD) unless otherwise noted.

## Results

Sex Ratio and LJFL.—Of the 645 specimens of blue marlin for which LJFL and sex data were available, 174 were identified as female and 471 were male. Sex ratios varied substantially among regions (Table 1). In Region I, specimens were collected from July to September, and all 100 individuals were females. In Region II, specimens were collected from September to November, and the sex ratio of the 54 individuals did not differ significantly from 1:1 (females:males = 28:26; Chi-square test: $\chi^2 = 0.074$, $P = 0.79$). In Region III, the sex ratio of the 473 individuals was strongly biased toward males (females:males = 34:439; $\chi^2 = 346.776$, $P < 0.01$) during the months in

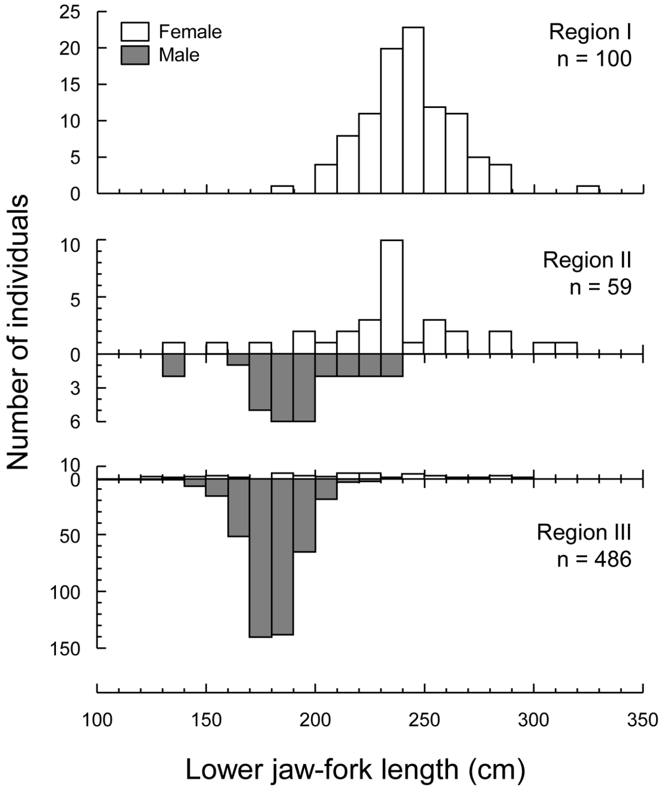


Figure 3. Lower jaw–fork length frequency distributions of female and male blue marlin, *Makaira nigricans*, in three different regions of the North Pacific Ocean.

which sufficient samples were obtained. Few juvenile females (n = 12) and males (n = 6) were observed in Regions II and III.

The mean LJFL of females (239 ± 26 cm, n = 162) was significantly larger than that of males (180 ± 13 cm, n = 465) excluding juvenile data (Welch's t-test: $t = 27.82$, $P < 0.01$; Fig. 3). The mean LJFL of females was significantly larger in Region I (243 ± 21 cm, n = 100) than in Region III (228 ± 32 cm, n = 34; $t = 2.949$, $P < 0.05$), and that of Region II (241 ± 29 cm, n = 28) did not differ significantly from Regions I ($t = 0.317$, $P = 0.75$) or III ($t = 2.029$, $P = 0.09$). The mean LJFL of males in Region II (194 ± 20 cm, n = 26) was significantly larger than in Region III (180 ± 12 cm, n = 439; $t = 3.621$, $P < 0.01$).

Reproductive Condition.—Excluding juvenile data, the histology of 152 female ovaries indicated that reproductive condition differed among samples collected in different regions (Table 2). No maturing or spawning females were observed in Regions I and II. However, one maturing (3%) and seven spawning (24%) females (184–287 cm LJFL) were observed in Region III in January, May, June, and October. All of these females were caught in the eastern area of Region III (Fig. 4). Postovulatory follicles were observed in ovaries in January and May (one specimen each), and hydrated oocytes were observed from another specimen in May. Immature stage ovaries were found only in juvenile female. Of the 301 testes examined, 232 (141–220 cm LJFL)

Table 2. Gonad condition of female and male blue marlin, *Makaira nigricans*, in the three different regions of the North Pacific Ocean. Only females were collected in Region I.

| | | Female | | | Male | |
|---|---|---|---|---|---|---|
| Area | Month | Inactive | Maturing | Spawning | Inactive | Active |
| Region I | | | | | | |
| | July | 48 | 0 | 0 | – | – |
| | August | 13 | 0 | 0 | – | – |
| | September | 36 | 0 | 0 | – | – |
| | Subtotal | 97 | 0 | 0 | – | – |
| Region II | | | | | | |
| | September | 5 | 0 | 0 | 0 | 5 |
| | October | 14 | 0 | 0 | 2 | 2 |
| | November | 7 | 0 | 0 | 2 | 1 |
| | Subtotal | 26 | 0 | 0 | 4 | 8 |
| Region III | | | | | | |
| | January | 1 | 0 | 1 | 16 | 39 |
| | February | 8 | 0 | 0 | 26 | 62 |
| | April | 1 | 0 | 0 | 6 | 5 |
| | May | 4 | 0 | 5 | 12 | 88 |
| | June | 4 | 1 | 0 | 2 | 17 |
| | July | 1 | 0 | 0 | 1 | 2 |
| | October | 1 | 0 | 1 | 0 | 0 |
| | November | 1 | 0 | 0 | 2 | 11 |
| | Subtotal | 21 | 1 | 7 | 65 | 224 |

exuded milt. The frequency of actively spawning males was high in both Regions II (67%) and III (78%).

Stomach Contents.—We examined a total of 484 blue marlin stomachs; 10% of these were empty. In total, 1942 individual prey items weighing 107.6 kg were recorded from 437 blue marlin stomachs. The prey species consisted of 12 cephalopod (two additional taxa remain unidentified and may be included in these 12), three crustacean, and 39 teleost species (five additional taxa remain unidentified, Table 3). In Region I, *Scomber* spp. (including *Scomber japonicus* Houttuyn, 1782 and *Scomber australasicus* Cuvier, 1831; %W = 34%) were the most abundant by weight, followed by *Auxis* spp. [including *Auxis thazard* (Lacépède, 1800) and *Auxis rochei* (Risso, 1810); 24%] and *Coryphaena hippurus* Linnaeus, 1758 (23%). In addition to *Scomber* spp. and *Auxis* spp., *Engraulis japonicus* Temminck and Schlegel, 1846 (%N = 20%, %F = 21%) and *Todarodes pacificus* (Steenstrup, 1880) (18%, 13%) were also abundant in number and frequency of occurrence in this region. In Region II, *Katsuwonus pelamis* (Linnaeus, 1758) (%W = 63%) was the most abundant by weight, followed by *Ranzania laevis* (Pennant, 1776) (10%) and *Alepisaurus ferox* Lowe, 1833 (8%). In this region, *Gempylus serpens* Cuvier, 1829 (%N = 16%, %F = 23%) was also abundant in number and frequency of occurrence in addition to the above three taxa. In Region III, *K. pelamis* (%W = 61%) was the most abundant by weight, and the large diversity of prey items resulted in no other important prey taxa exceeding 5% by weight. In addition to *K. pelamis,* species of Teuthoidea (%N = 16%, %F = 32%) and *G. serpens* (21%, 28%) were abundant in number and frequency of occurrence in this region.

Table 3. List of prey items found in the stomachs of blue marlin, *Makaira nigricans*, sampled in the three different regions of the North Pacific Ocean. Prey length is given in mantle length for Cephalopoda, total length for Crustacea, and standard length for Teleostei. %F: frequency of occurrence, %N: number, %W: weight, %IRI: index of relative importance.

| | Region I (n = 90) | | | | Region II (n = 39) | | | | Region III (n = 308) | | | | Length (mm) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Prey taxon | %F | %N | %W | %IRI | %F | %N | %W | %IRI | %F | %N | %W | %IRI | Min | Max |
| Cephalopoda | | | | | | | | | | | | | | |
| *Loligo bleekeri* | 1.1 | 0.5 | 0.2 | 0.0 | – | – | – | – | – | – | – | – | 110 | 160 |
| *Ancistrocheirus lesueuri* | – | – | – | – | – | – | – | – | 0.3 | 0.1 | 0.0 | 0.0 | | |
| *Onychoteuthis banksii* | – | – | – | – | 2.6 | 0.9 | 0.3 | 0.1 | 0.6 | 0.2 | 0.1 | 0.0 | 80 | 95 |
| *Moroteuthis loennbergi* | – | – | – | – | – | – | – | – | 0.3 | 0.1 | 0.1 | 0.0 | 100 | |
| *Todarodes pacificus* | 13.3 | 17.6 | 4.5 | 8.3 | – | – | – | – | – | – | – | – | 110 | 200 |
| *Ommastrephes bartrami* | 6.7 | 0.9 | 0.7 | 0.3 | – | – | – | – | – | – | – | – | 100 | 197 |
| *Sthenoteuthis oualaniensis* | – | – | – | – | – | – | – | – | 1.0 | 0.4 | 0.5 | 0.0 | 70 | 125 |
| *Hyaloteuthis pelagica* | – | – | – | – | – | – | – | – | 0.6 | 0.3 | 0.1 | 0.0 | 57 | 73 |
| Ommastrephidae spp. | 1.1 | 0.2 | 0.0 | 0.0 | 5.1 | 2.6 | 0.9 | 0.4 | 9.4 | 5.0 | 2.7 | 1.7 | 55 | 112 |
| *Thysanoteuthis rhombus* | – | – | – | – | – | – | – | – | 0.3 | 0.1 | 0.2 | 0.0 | 132 | |
| *Mastigoteuthis cordiformis* | – | – | – | – | – | – | – | – | 0.3 | 0.1 | 0.0 | 0.0 | | |
| Teuthoidea spp. | – | – | – | – | 5.1 | 1.7 | 0.1 | 0.2 | 31.8 | 16.2 | 3.3 | 14.6 | 30 | 113 |
| Octopoda spp. | – | – | – | – | 10.3 | 5.1 | 0.1 | 1.2 | 6.2 | 2.7 | 0.8 | 0.5 | 10 | 82 |
| *Argonauta* spp. | – | – | – | – | 2.6 | 0.9 | 0.0 | 0.0 | 4.2 | 1.5 | 1.1 | 0.3 | 10 | 90 |
| Crustacea | | | | | | | | | | | | | | |
| *Phrosina semilunata* | – | – | – | – | 2.6 | 1.7 | 0.0 | 0.1 | – | – | – | – | 18 | 19 |
| Amphipoda spp. | – | – | – | – | 5.1 | 5.1 | 0.0 | 0.6 | 0.3 | 0.2 | 0.0 | 0.0 | 12 | 17 |
| Caridea spp. | – | – | – | – | 2.6 | 0.9 | 0.0 | 0.1 | 0.3 | 0.1 | 0.0 | 0.0 | 20 | 74 |
| Teleostei | | | | | | | | | | | | | | |
| *Etrumeus teres* | 3.3 | 2.1 | 0.4 | 0.2 | – | – | – | – | – | – | – | – | 98 | 160 |
| *Engraulis japonicus* | 21.1 | 20.2 | 1.9 | 13.2 | – | – | – | – | – | – | – | – | 90 | 136 |
| *Alepisaurus ferox* | – | – | – | – | 17.9 | 6.0 | 8.2 | 5.7 | 2.6 | 0.7 | 3.3 | 0.2 | 243 | 1250 |
| *Lophotus capellei* | – | – | – | – | – | – | – | – | 1.6 | 0.4 | 1.2 | 0.1 | 320 | 500 |
| *Myripristis* sp. | – | – | – | – | – | – | – | – | 0.3 | 0.1 | 0.0 | 0.0 | 54 | |

Table 3. Continued.

| | Region I (n = 90) | | | | Region II (n = 39) | | | | Region III (n = 308) | | | | Length (mm) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Prey taxon | %F | %N | %W | %IRI | %F | %N | %W | %IRI | %F | %N | %W | %IRI | Min | Max |
| Hemiramphidae sp. | – | – | – | – | – | – | – | – | 0.3 | 0.1 | 0.0 | 0.0 | 80 | |
| Exocoetidae spp. | – | – | – | – | – | – | – | – | 0.6 | 0.2 | 0.3 | 0.0 | 200 | |
| Belonidae sp. | – | – | – | – | – | – | – | – | 0.3 | 0.3 | 0.2 | 0.0 | | |
| *Cololabis saira* | 1.1 | 0.5 | 0.3 | 0.0 | – | – | – | – | – | – | – | – | 260 | 270 |
| *Coryphaena hippurus* | 13.3 | 2.8 | 23.0 | 9.7 | 2.6 | 0.9 | 1.9 | 0.2 | 1.6 | 0.5 | 0.4 | 0.0 | 60 | 670 |
| *Coryphaena equiselis* | – | – | – | – | – | – | – | – | 2.3 | 0.6 | 2.3 | 0.2 | 150 | 277 |
| *Decapterus* spp. | 8.9 | 1.7 | 1.3 | 0.7 | 5.1 | 1.7 | 2.4 | 0.5 | 3.9 | 1.0 | 1.8 | 0.3 | 138 | 265 |
| Carangidae sp. | 1.1 | 0.2 | 0.2 | 0.0 | – | – | – | – | 0.6 | 0.3 | 0.1 | 0.0 | 120 | |
| *Brama* spp. | – | – | – | – | – | – | – | – | 9.1 | 3.0 | 1.0 | 0.9 | 40 | 115 |
| Nomeidae sp. | 1.1 | 0.2 | 0.8 | 0.0 | – | – | – | – | – | – | – | – | 310 | |
| *Pseudoscopelus sagamianus* | – | – | – | – | – | – | – | – | 1.0 | 0.3 | 0.0 | 0.0 | 48 | 86 |
| *Naso unicornis* | – | – | – | – | – | – | – | – | 2.6 | 1.7 | 0.3 | 0.1 | 50 | 65 |
| Istiophoridae sp. | – | – | – | – | – | – | – | – | 7.1 | 1.9 | 0.4 | 0.4 | 180 | |
| *Makaira nigricans* | – | – | – | – | 2.6 | 0.9 | 0.0 | 0.1 | 2.6 | 1.4 | 0.8 | 0.1 | 160 | 212 |
| *Xiphias gladius* | – | – | – | – | – | – | – | – | 0.6 | 0.2 | 0.0 | 0.0 | 173 | 220 |
| *Lepidocybium flavobrunneum* | 1.1 | 0.2 | 0.4 | 0.0 | – | – | – | – | – | – | – | – | 244 | |
| *Ruvettus pretiosus* | – | – | – | – | – | – | – | – | 0.3 | 0.1 | 0.0 | 0.0 | 89 | |
| *Gempylus serpens* | – | – | – | – | 23.1 | 16.2 | 1.7 | 9.3 | 27.6 | 20.5 | 4.9 | 16.6 | 150 | 470 |
| *Nesiarchus nasutus* | – | – | – | – | – | – | – | – | 0.3 | 0.1 | 0.0 | 0.0 | 110 | |
| Gempylidae spp. | – | – | – | – | 15.4 | 6.0 | 0.7 | 2.3 | 12.0 | 7.7 | 0.8 | 2.4 | 150 | 300 |
| *Scomber japonicus* | 18.9 | 15.6 | 18.1 | 18.0 | – | – | – | – | – | – | – | – | 190 | 285 |
| *Scomber australasicus* | 2.2 | 2.5 | 5.1 | 0.5 | – | – | – | – | – | – | – | – | 243 | 295 |
| *Scomber* spp. | 22.2 | 9.2 | 11.0 | 12.7 | – | – | – | – | 0.3 | 0.1 | 0.5 | 0.0 | 210 | 281 |
| *Auxis thazard* | 2.2 | 0.3 | 0.4 | 0.0 | – | – | – | – | 0.3 | 0.1 | 0.2 | 0.0 | 260 | |
| *Auxis rochei* | 28.9 | 12.0 | 19.4 | 25.6 | – | – | – | – | – | – | – | – | 140 | 339 |
| *Auxis* spp. | 16.7 | 3.4 | 4.4 | 3.6 | 10.3 | 4.3 | 6.6 | 2.5 | 0.6 | 0.2 | 0.8 | 0.0 | 140 | 300 |

Table 3. Continued.

| | Region I (n = 90) | | | | Region II (n = 39) | | | | Region III (n = 308) | | | | Length (mm) | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Prey taxon | %F | %N | %W | %IRI | %F | %N | %W | %IRI | %F | %N | %W | %IRI | Min | Max |
| *Katsuwonus pelamis* | 8.9 | 2.6 | 3.4 | 1.5 | 33.3 | 14.5 | 63.3 | 58.2 | 27.9 | 10.9 | 60.6 | 47.1 | 89 | 483 |
| *Thunnus thynnus* | 4.4 | 0.9 | 1.3 | 0.3 | – | – | – | – | – | – | – | – | 140 | 240 |
| *Thunnus albacares* | – | – | – | – | 5.1 | 1.7 | 1.6 | 0.4 | 0.3 | 0.1 | 0.2 | 0.0 | 145 | 226 |
| *Thunnus obesus* | – | – | – | – | – | – | – | – | 0.3 | 0.1 | 0.2 | 0.0 | 162 | |
| Scombridae spp. | 4.4 | 1.1 | 0.8 | 0.2 | 5.1 | 1.7 | 0.2 | 0.2 | 10.1 | 2.7 | 2.0 | 1.1 | | |
| Balistidae spp. | – | – | – | – | – | – | – | – | 1.3 | 0.4 | 0.0 | 0.0 | 38 | 41 |
| *Aluterus scriptus* | – | – | – | – | – | – | – | – | 0.3 | 0.1 | 1.5 | 0.0 | 360 | |
| *Lagocephalus lagocephalus* | – | – | – | – | – | – | – | – | 3.2 | 1.0 | 0.4 | 0.1 | 50 | 148 |
| Tetraodontidae sp. | 1.1 | 0.2 | 0.0 | 0.0 | – | – | – | – | – | – | – | – | | |
| *Diodon* sp. | – | – | – | – | – | – | – | – | 0.6 | 0.2 | 0.1 | 0.0 | | |
| *Masturus lanceolatus* | – | – | – | – | – | – | – | – | 2.6 | 0.9 | 0.5 | 0.1 | 25 | 138 |
| *Ranzania laevis* | – | – | – | – | 20.5 | 16.2 | 10.1 | 12.1 | 2.9 | 0.9 | 3.6 | 0.3 | 60 | 320 |
| Unidentified Teleostei | 22.2 | 5.4 | 2.5 | 4.9 | 20.5 | 11.1 | 1.8 | 5.9 | 31.5 | 14.5 | 2.8 | 12.9 | 5 | 180 |

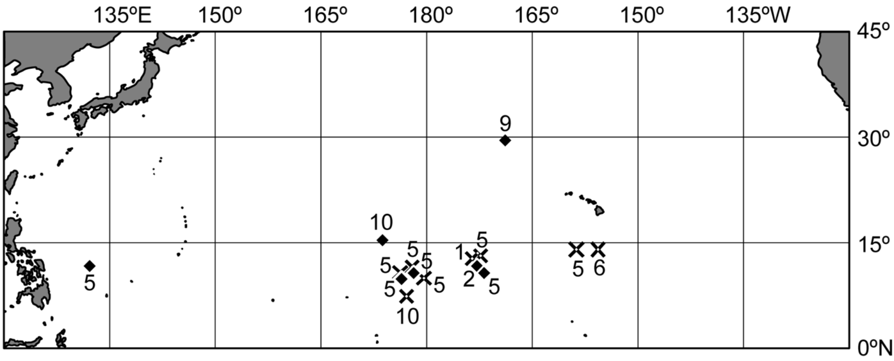


Figure 4. Location of eight maturing or spawning female blue marlin, *Makaira nigricans* (crosses), and 16 cannibalized juveniles (~1 mo old) from the stomachs of nine blue marlin in seven locations (diamonds). Numbers by symbols indicate month of collection.

Cannibalism of juvenile blue marlin (160–212 mm) was observed. One specimen was found in Region II in September, and 16 specimens were found in eight stomachs from six localities in western and eastern areas of Region III in February (n = 1), May (n = 5), and October (n = 10; Fig. 4).

Teuthoidea and Thunnini (*Auxis, Katsuwonus,* and *Thunnus*) were consistently observed in stomachs of blue marlin collected using both types of fishing gear (n = 125 and 132 via lure trolling; n = 269 and 191 via longline). Digestive classes did not differ significantly between samples from the two types of gear for either Teuthoidea (Mann–Whitney U-test: U = 17,501, P = 0.50) or Thunnini (U = 13,665, P = 0.18), indicating that the digestive conditions were similar between samples from both types of gear. Therefore, values of stomach content weight and SCI can be compared using samples from both types of gear.

Feeding Intensity.—Single-prey weight (digestive classes 1 and 2) in Region I (median = 90 g, n = 202) was significantly heavier than values in Region II (35 g, n = 27; Mann-Whitney U-test: P < 0.01) and Region III (17 g, n = 197; P < 0.01; Fig. 5). The mean number of prey per stomach was 7.24 in Region I, which was nearly double the values in Regions II (3.00) and Region III (3.81).

Stomach content indices of females were significantly higher in Region I (mean ± SE = 0.47 ± 0.06, median = 0.34, n = 94) than in Region II (0.13 ± 0.05, 0.06, n = 18; Mann-Whitney U-test: P < 0.01) and Region III (0.20 ± 0.10, 0.01, n = 25; P < 0.01; Fig. 6). The stomach content indices of males were significantly higher in Region II (0.64 ± 0.26, 0.34, n = 22) than in Region III (0.18 ± 0.02, 0.06, n = 311; P < 0.01).

## Discussion

Spawning and Feeding.—The occurrence of maturing and spawning females in January, May, June, and October suggests that spawning may occur throughout the year in Region III (4°N–21°N, 131°E–154°W). Neither maturing (3%) nor spawning (24%) females were abundant in Region III; low frequency of occurrence of spawning females is common for blue marlin (Shimose et al. 2009, Sun et al. 2009). Based on a juvenile aging study in the Atlantic (Prince et al. 1991), the cannibalized juvenile blue

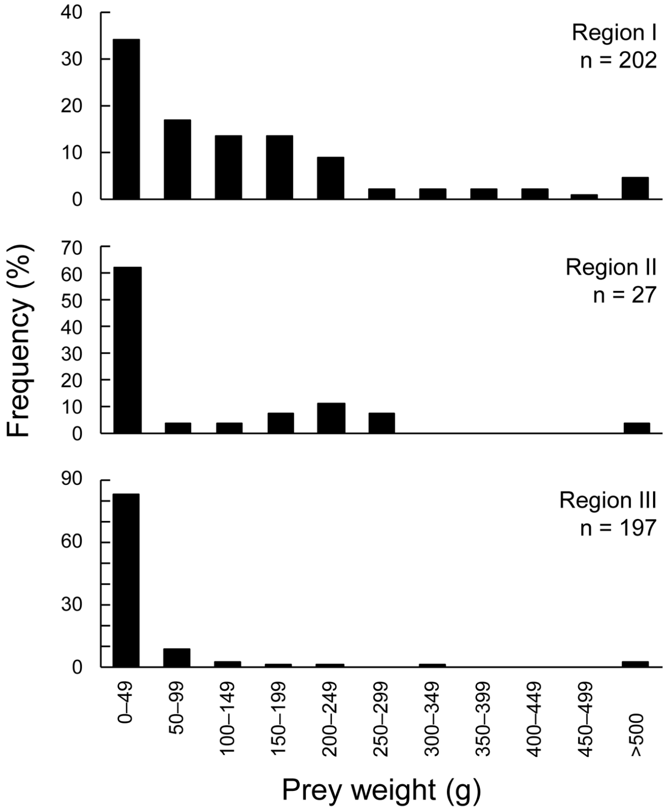


Figure 5. Prey weight frequency distribution of blue marlin, *Makaira nigricans*, in three different regions of the North Pacific Ocean. Note *y*-axis scales vary among plots.

marlin (160–212 mm) found in stomachs in February, May, and October during the present study were likely ~1 mo of age, indicating that spawning occurred in January, April, and September in Region III. Although the spawning females were not abundant in the present study, their appearance in the samples is consistent with the hypothesis that spawning occurs during all seasons in Region III, as shown by previous larval-occurrence data (Nishikawa et al. 1985). The spawning season in the western North Pacific (16°N–23°N, 115°E–135°E) and at Yonaguni Island (24°27′N, 122°57′E) in southwestern Japan has been estimated to occur from May to September (Shimose et al. 2009, Sun et al. 2009). These spawning areas and Region III are located within a known extensive spawning ground (Nakamura 1983, Nishikawa et al. 1985). Spawning seasonality in the slightly higher latitude area (16°N–24°N) from May to September (Shimose et al. 2009, Sun et al. 2009) may be caused by the northward expansion of the primary spawning area during those months. However, neither evidence of spawning by females nor the occurrence of male individuals was observed in Region I (33°N–36°N, 135°E–140°E). Blue marlin is the most tropical species of billfish (Nakamura 1985), and the restricted spawning area reflects the tropical origin of these fishes (Boyce et al. 2008).

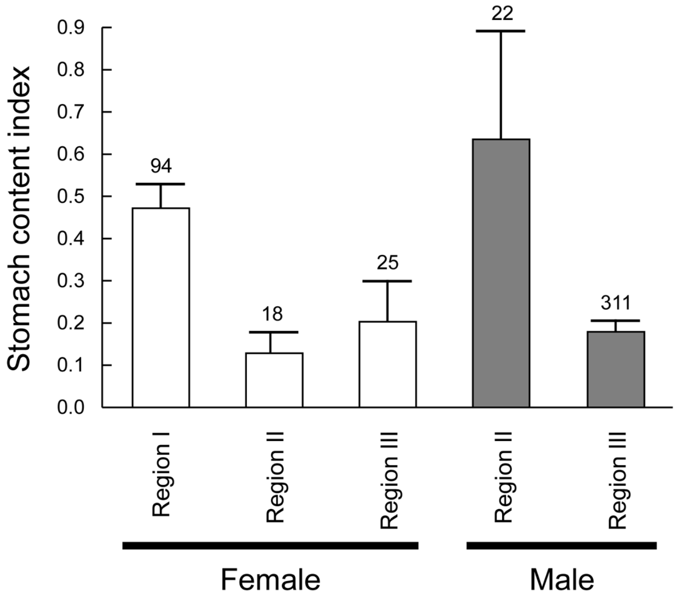


Figure 6. Mean (± SE) stomach content index of female and male blue marlin, *Makaira nigricans*, in the three different regions of the North Pacific Ocean. Numbers above are sample sizes.

Blue marlin fed mainly on teleosts, and prey composition changed across regions. Important prey items included scombrid fishes (e.g., *Scomber* spp. and *Auxis* spp.) and *C. hippurus*, both of which have relatively large body masses. These two prey species contributed to the high SCI of blue marlin in Region I (33°N–36°N, 135°E–140°E). Blue marlin prey items from fish in Region I were not only larger but also more abundant than those in stomachs of fish in the other two regions. Consequently, many large prey items contributed to the high SCI in Region I (mean = 0.47); this value is similar to that observed for blue marlin around Yonaguni Island (mean = 0.43, considering empty stomachs as SCI = 0.00; Shimose et al. 2006). During the summer in the mid-1980s in Region I, *Sardinops melanostictus* (Schlegel, 1846) was the second-most numerous prey of blue marlin, after *Scomber* spp. (Yatomi 1995). However, stocks of *S. melanostictus* around Japan declined after 1988 (Noto and Yasuda 1999), which may explain why this species was not recorded in the present study. Alternatively, *E. japonicus* and *T. pacificus* occurred frequently in blue marlin stomachs during the present study period, which may imply species replacement of *S. melanostictus* due to the regime shift from cold to warm during the 1980s (Kawasaki 1983, Sakurai 2007). Although the dominant nektonic species have shifted over longer time scales (*S. melanostictus* in the mid-1980s vs *E. japonicus* and *T. pacificus* between 2003 and 2009), based on stomach contents, suitably sized prey for blue marlin appear to have been abundant in Region I.

*Katsuwonus pelamis* was the most important prey in Regions II (8°N–32°N, 171°E–140°W) and III (4°N–21°N, 131°W–154°W). These results were similar to diet studies in Hawaii (Brock 1984) and around Yonaguni Island (Shimose et al. 2006). This species is likely an important prey item for blue marlin because it is distributed throughout the Pacific Ocean even in areas of low primary production (Longhurst et al. 1995, Lehodey et al. 1997), whereas other suitable prey items may not have been abundant in Regions II and III. Scombridae includes both neritic (*Scomber* spp. and *Auxis* spp.) and oceanic (*K. pelamis*) species (Collette and Nauen 1983), and these fish constitute important blue marlin prey in both coastal (Region I) and oceanic (Regions II and III) regions. In addition to *K. pelamis,* small deep-dwelling prey items such as *Teuthoidea* spp. and *G. serpens* were consumed frequently in Regions II and III. Compared with larger individuals, smaller blue marlin tend to feed on smaller and deep-dwelling prey items (e.g., Ommastrephidae and Gempylidae; Shimose et al. 2006). Striped marlin are known to dive to feed on deep-dwelling prey items in areas where other suitable prey items are scarce (Shimose et al. 2010). The frequent occurrence of small, deep-dwelling prey items in Regions II and III suggests that more preferable prey items were rare in these areas.

Differential Migration Between the Sexes.—The present study suggests that the known north–south migration of blue marlin in the western North Pacific Ocean (Anraku and Yabuta 1959) is strongly related to their reproductive and feeding activities in addition to water temperature. Blue marlin migrate to Region I only from May to November, with their occurrence there peaking in August and September (Yatomi 1995). At Yonaguni Island (24°27´N, 122°57´E), both female and male blue marlin are abundant from January to September, with evidence of active feeding, and few females spawn from May to September (Shimose et al. 2006, 2009). Taken together, data from Regions I and III and from around Yonaguni Island (Shimose et al. 2006, 2009), where sufficient data or information are available in all seasons, reveal that blue marlin migrate to different areas for probably different purposes (Fig. 7): (1) in Region III, males dominate throughout the year, and spawning occurs; (2) some females and males migrate around Yonaguni Island from lower latitude areas for feeding during January–September; (3) the spawning area expands to around Yonaguni Island during May–September; (4) some females migrate to Region I for feeding during July–September; and (5) females and males migrate to Region III during October–December. Male blue marlin remain in Region III throughout the year, where spawning grounds exist but prey items are scarce. Once mature, male blue marlin maintain sufficient spermatozoa in their testes for spawning year round (de Sylva and Breder 1997, Shimose et al. 2009). Body size may not be a key factor to spawning success for male blue marlin, which are typically much smaller than females. Therefore, waiting for females that are in spawning condition while forgoing the feeding migration is likely an efficient reproductive strategy for males. On the other hand, female blue marlin are larger and migrate to Region I for feeding during the summer season. This region is far from the spawning grounds, but prey items appear to be abundant as indicated by stomach contents of fish collected in this region. Females expend their body energy on reproduction from May to September in the western North Pacific (Shimose et al. 2009, Sun et al. 2009), and post-spawning females may migrate to Region I during July–September to recover their body condition. From October to December, female blue marlin leave Region I (Yatomi 1995), and catch numbers decrease at Yonaguni Island (Shimose et al. 2009). Blue marlin prefer water temperatures ~26–27 °C (Boyce et al. 2008), and their distribution may be confined to areas warmer than the 24 °C surface isotherm (Nakamura 1985). Females and males may return to Region III to avoid lower water temperatures, both

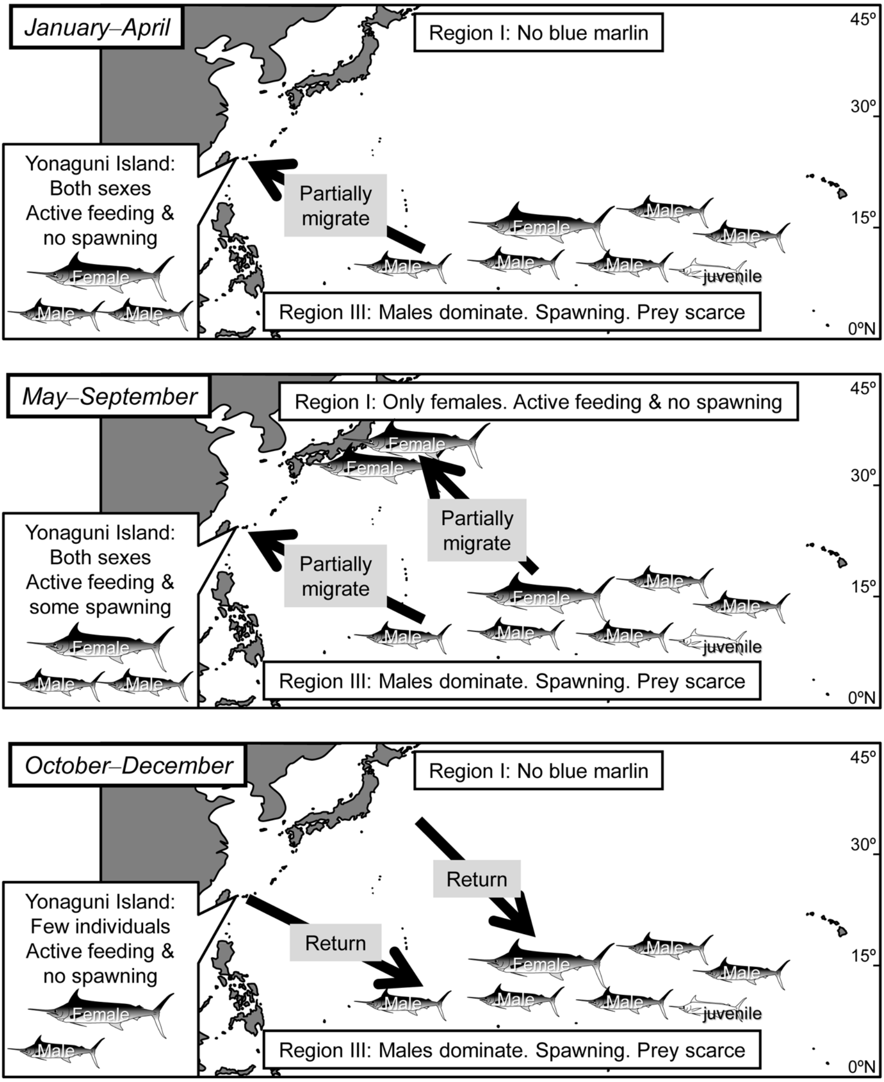


Figure 7. Schematic summary of the spawning and feeding of blue marlin, *Makaira nigricans*, in the North Pacific Ocean. Data for Yonaguni Island are from previous studies (Shimose et al. 2006, 2009). Arrows show possible migration of blue marlin.

along the coast and at higher latitudes; thus, the return may not represent a spawning migration per se.

The sex ratio of blue marlin changes across areas and seasons (Kume and Joseph 1969, Hopper 1990, González-Armas et al. 2006). The equal sex ratio in Region II was more similar to that observed at Yonaguni Island (females:males = 2:1, Shimose et al. 2009) than the other regions. These sex ratios were intermediate between Region I (only females) and Region III (males dominate). Striped marlin, *Kajikia audax* (Philippi, 1887), which is smaller than blue marlin, were also landed in Region I (T Shimose, pers obs), indicating that there was unlikely a gear bias toward large female blue marlin in Region I. These results suggest that females tend to migrate farther from the spawning area than do males, causing a latitudinal cline in sex ratio. Clines in sex ratios have also been observed for swordfish in the western Atlantic (Arocha and Lee 1996) and in the area north of Hawaii (DeMartini et al. 2000). Furthermore, sex ratios are strongly biased toward females for swordfish in New Zealand waters (Young et al. 2003) and for black marlin *Istiompax indica* (Cuvier, 1832) at Yonaguni Island (Shimose et al. 2008). All of these billfish species exhibit sexual dimorphism in which females are much larger than males (Nakamura 1983), and female-biased areas are located far from the spawning area for each species (Young et al. 2003, Shimose et al. 2008). The sexual dimorphism of body size may allow only large females to migrate far from the spawning ground. Whether this migration occurs for purposes of growth or to recover energy lost from spawning is not clear. However, sexual differences in migration are a key issue for understanding the reproductive strategy and sexual dimorphism in body size of oceanic animals (Ichii et al. 2009). The occurrence of juvenile female (< 180 cm) and male (< 140 cm) blue marlin was restricted to Regions II and III, and blue marlin of these sizes are thought to be < 1 yr of age (Prince et al. 1991). These findings suggest that these areas are nursery grounds, which appear to occur over a slightly broader geographical area than the spawning ground in Region III. Blue marlin begin to migrate to distant but rich feeding grounds only after they attained a sufficiently large size.

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